Assignment 4

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Main Memory

Address	Data				Address	Data			
0x0000	aa	bb	cc	dd	0x011C	ab	bc	ab	bc
0x0004	12	bc	12	bc	0x0120	ab	bb	ab	bb
0x0008	ab	bb	cd	bb	0x0124	ab	bc	ab	bc
0x000C	1b	bc	1c	bc	0x0128	ab	bb	ab	bb
0x0010	ab	bb	cb	bb	0x012C	ab	bc	ab	bc
0x0014	1b	bc	1b	be	0x0130	ab	bc	ab	bc
0x0018	12	bc	12	bf	0x0134	ab	bb	ab	bb
0x001C	1b	bb	1b	bd	0x0138	ab	bc	ab	bc
0x0020	1b	bc	1b	b1	0x013C	ab	bb	ab	bb
0x0024	1b	bb	1b	bb	0x0140	ab	bc	ab	bc
0x0028	1b	bc	11	bc	0x0144	ab	bc	ab	bc
0x002C	12	bc	1b	bb	0x0148	ab	bb	ab	bb
0x0030	1b	bb	1b	be	0x014C	ab	bc	ab	bc
0x0034	1b	bc	13	bf	0x0150	ab	bb	ab	bb
0x0038	1b	bb	1b	bd	0x0154	ab	bc	ab	bc
0x003C	1b	bc	1b	b2	0x0158	ab	bc	ab	bc
0x0040	11 01	bc 01	1b 01	bb 01	0x015C	ab	bb	ab	bb
0x0044	1b	bb	1b	bc	0x0160	ab	bc	ab	bc
0x0048	1b	bc	1b	bb	0x0164	ab	bb	ab	bb
0x004C	1b	bb	1b	be	0x0168	ab	bc	ab	bc
0x0050	1b	bc	14	bf	0x016C	ab	bc	ab	bc
0x0054	ab	bc	1b	bd	0x0170	ab	bb	ab	bb
0x0058	1b	bb	1b	b3	0x0174	ab	bc	ab	bc
0x005C	ab	bc	1b	bc	0x0178	ab	bb	ab	bb
0x0060	1b	bb	1b	bb	0x017C	ab	bc	ab	bc
0x0064	ab 98	bc 11	1b 22	bc 23	0x0180	ab	bc	ab	bc
0x0068	ab	bc	ab	bc	0x0184	ab	bb	ab	bb
0x006C	ab	bb	1b	bb	0x0188	ab	bc	ab	bc
0x0070	ab	bc	ab	bc	0x018C	ab	bb	ab	bb
0x0074	ab	bb	1b	bb	0x0190	ab	bc	ab	bc
0x0078	ab	bc	ac	bc	0x0194	ab	bc	ab	bc
0x007C	ab	bc	1b	bc	0x0198	ab	bb	ab	bb
0x0080	ab	bb	ac	bb	0x019C	ab	bc	ab	bc
0x0084	ab	bc	1c	bc	0x01A0	ab	bb	ab	bb
0x0088	ab	bb	ab	bb	0x01A4	ab	bc	ab	bc

0x008C	ab	bc	1c	bc	0x01A8	ab	bc	ab	bc
0x0090	ab	bc	ab	bc	0x01AC	ab	bb	ab	bb
0x0094	ab cc	bb cc	1c cc	bb cc	0x01B0	ab	bc	ab	bc
0x0098	ab	bc	ac	bc	0x01B4	ab dd	bb dd	ab dd	bb dd
0x009C	ab	bb	1b	bb	0x01B8	ab	bc	ab	bc
0x00A0	ab 02	bc 02	ac 02	bc 02	0x01BC	ab	bc	ab	bc
0x00A4	ab ab	bc bc	1b 1b	bc bc	0x01C0	ab	bb	ab	bb
0x00A8	ab	bb	ac	bb	0x01C4	ab	bc	ab	bc
0x00AC	ab	bc	1c	bc	0x01C8	ab	bb	ab	bb
0x00B0	ab 15	bb bb	ab 16	bb bb	0x01CC	ab	bc	ab	bc
0x00B4	ab	bc	1c	bc	0x01D0	ab	bc	ab	bc
0x00B8	ab	bc	ab	bc	0x01D4	ab	bb	ab	bb
0x00BC	ab	bb	1c	bb	0x01D8	ab	bc	ab	bc
0x00C0	ab	bc	ac	bc	0x01DC	ab	bb	ab	bb
0x00C4	ab	bb	1b	bb	0x01E0	2c	bc	ab	bc
0x00C8	ab	bc	ac	bc	0x01E4	ab	bc	ab	bc
0x00CC	ab	bc	1b	bc	0x01E8	2b	bb	ab	bb
0x00D0	ab	bb	ac	bb	0x01EC	ab	bc	ab	bc
0x00D4	fb	bc	1c	bc	0x01F0	2b e0	bb e0	ab 0e	bb 0e
0x00D8	ab	bb	ab	bb	0x01F4	ab	bc	ab	bc
0x00DC	fb	bc	1c	bc	0x01F8	ab	bc	ab	bc
→ 0x00E0	ab	bc	ab	bc	0x01FC	ab	bb	ab	bb
0x00E4	fb	bb	1c	bb	0x0200	ab	bc	ab	bc
0x00E8	ab	bc	ab	bc	0x0204	ab	bb	ab	bb
0x00EC	fb	bb	1b	bb	0x0208	ab	bc	ab	bc
0x00F0	ab	bc	ab	bc	0x020C	ab	bc	ab	bc
0x00F4	ab	bc	ab	bc	0x0210	1c	bb	ab	bb
0x00F8	ab	bb	ab	bb	0x0214	ab	bc	ab	bc
0x00FC	ab	bc	ab	bc	0x0218	1b	bb	ab	bb
0x0100	ab	bb	ab	ff	0x021C	ab	bc	ab	bc
0x0104	ab	bc	ab	bc	0x0220	ab	bc	ab	bc
0x0108	ab	bc	ab	bc	0x0224	ab	bb	ab	bb
0x010C	ab	bb	ab	bb	0x0228	ab	bc	ab	bc
0x0110	ab	bc	ab	bc	0x022C	ab	bb	ab	bb
0x0114	ab	bb	ab	bb	0x0230	ab	bc	ab	bc
0x0118	ab	bc	ab	bc	0x0234	ff	ff	ff	ff

Memory access sequence:

Op	Address	Addresses in binary	Data (Hex)	Hit/Miss (a)	Hit/Miss (b)	Hit/Miss (c)
LD	0x00E0	0000 0000 1110 0000	ab bc ab bc	M	H	M
LD	0x0050	0000 0000 0101 0000	1b bc 14 bf	M	M	M
ST	0x0040	0000 0000 0100 0000	01 01 01 01	M (Allocate)	M(A)	M(A)
ST	0x00A3	0000 0000 1010 0011	02 02 02 02	M(A)	M(A)	M(A)
LD	0x01E7	0000 0001 1110 0111	Ab bc ab bc	M	M	M
ST	0x01B4	0000 0001 1011 0100	dd dd dd dd	M(A)	M(A)	M(A)
ST	0x01F0	0000 0001 1111 0000	e0 0	M(A)	M(A)	M(A)
LD	0x0130	0000 0001 0011 0000	ab bc ab bc	M	M	M
LD	0x0064	0000 0000 0110 0100	ab bc 1b bc	M	M	M
ST	0x00B0	0000 0000 1011 0000	15 bb 1	M(A)	M(A)	M(A)
ST	0x0066	0000 0000 0110 0110	3 11 2 3	H (Through)	HL Back)	H(B)
ST	0x0094	0000 0000 1001 0100	ee	M	M	M
LD	0x01E4	0000 0001 1110 0100	ab bc ab bc	M	H	H

Now consider three different 32-bytes ca s shown below. Assume that each of the four caches was used independently to facilitate mem y acce or the s ence above. For each cache type, assume that the cache is initially empty. Block- = 4 b s. Cac pacity=32 bytes. Note: If any byte requested from the proces or is not given in th ble b , just e 'g' as data at that address. Calculate **hit rate and miss rate**. For part b and c an se th tial val s of memory, not the updated values after part a.

Assume that the least-recently sed (LR schem is used where appropriate, where the least recently inserted or access d element replace on conf t. Also, when inserting an element into the cache, if there are multiple empty slots for one ind x, you should insert the new element into the (first available slot).

1]. Use the direct-mapped c e to f ata m ry access for the memory sequence above. You should fill in the binary form he Ta alues. y the final contents of the cache in the table below. nd com ute the hit and s rate. cache write hits use write through and for cache write m ses us te no a le [1 rks]

b = 4

C = 32

$B = \frac{32}{4} = 8$

N = 1

$S = \frac{B}{N} = 8$

$O.B = \log_2 b = 2$

$S.B = \log_2 S = 3$

$T.B = 16 - 2 - 3 = 11$ bits

$Hit\ \% = \frac{1}{13} \times 100 = 7.69\ \%$

$Miss\ \% = 100 - 7.69 = 92.3\ \%$

Set bits	V	Tag bits	Data			
			00	01	10	11
000	1	0000 0000 101	~~ab~~ ~~01~~ 02	~~bc~~ ~~01~~ 02	~~ab~~ ~~01~~ 02	~~bc~~ ~~01~~ 02
001	1	0000 0001 111	~~ab~~ ~~ab~~ ~~45~~ ab	~~bc~~ ~~bc~~ ~~11~~ bc	~~ab~~ ~~fb~~ ~~22~~ ab	~~bc~~ ~~bc~~ ~~33~~ bc
010	0					
011	0					
100	1	0000 0000 101	~~fb~~ ~~e0~~ ~~ab~~ 15	~~ba~~ ~~e0~~ ~~ab~~ bb	~~f4~~ ~~0e~~ ~~ab~~ 16	~~bf~~ ~~0e~~ ~~bc~~ bb
101	1	0000 0000 100	~~dd~~ cc	~~dd~~ cc	~~dd~~ cc	~~dd~~ cc
110	0					
111	0					

2]. Use 4-way associative cache to facilitate memory access for the memory sequence above. You should fill in the binary form of the Tag values. Show the final contents of the cache in the table below, and compute the

b=4 $S = \frac{B}{N} = 2$

C=32

B=8

N=4

O.B = 2

SB = 1

T.B = 16-2-1 = 13

Hit % = $\frac{2}{13} \times 100 = 15.38\%$

Miss % = 100 − 15.38 = 84.61%

hit rate and miss rate. For cache write hits use write back and for cache write misses use write allocate. You may add any column you find necessary. [15 marks]

Set bits	V	Tag bits	Dirty Bit	LRU	Data 00	Data 01	Data 10	Data 11
0	1	0000 0000 1110 0000 0001 1111 0	0	10	ab e0	abc e0	ab de	bc e0
	1	0000 0000 0101 0000 0001 0011 0	0	01	tb ab	bc bc	t4 ab	bf bc
	1	0000 0000 0000 0000 0000 1011 0	0	00	et 15	et bb	et 16	et bb
	1	0000 0000 1010 0	0	11	02	02	02	02
1	1	0000 0001 1110 0	0	00	ab	bc	ab	bc
	1	0000 0001 1011 0	0	11	dd	dd	dd	dd
	1	0000 0000 0110 0000 0000 1101 0	1	10	ab 98	bc 11	tb 22	bc 23

			Dirty	LRU				
		0000 0000 1001 0	0	00 01	cc	cc	cc	cc

3]. Use fully associative cache to facilitate memory access for the memory sequence above. You should fill in the binary form of the Tag values. Show the final contents of the cache in the table below, and compute the hit rate and miss rate. For cache write hits use write back and for cache write misses use write allocate. **You may add any column you find necessary. [15 marks]**

Dirt Bit	V	Tag bits	LRU	Data			
~~0~~ 1	1	~~0000 0000 1110 00~~ ~~0000 0000 0110 01~~ 0000 0000 0110 01	~~000 001~~ ~~010 101~~ ~~110 111~~ ~~000 001~~ 010	~~ab~~ ~~ac~~ 98	~~bc~~ ~~bc~~ 11	~~ab~~ ~~bb~~ 22	~~bc~~ ~~bc~~ 33
0	1	~~0000 0000 0101 00~~ 0000 0000 1011 00	~~000 001~~ ~~010 011~~ ~~100 101~~ ~~110 111~~ ~~000 001~~ ~~010 011~~	~~bb~~ 15	~~bc~~ bb	~~bb~~ 16	~~bf~~ bb
0	1	~~0000 0000 0100 00~~ 0000 0000 1001 01	~~001 001~~ ~~010 011~~ ~~100 101~~ ~~110~~ 011	~~cc~~ cc	~~cc~~ cc	~~cc~~ cc	~~cc~~ cc
0	1	~~0000 0000 1010 00~~	~~000 001~~ ~~010 011~~ ~~100 101~~ ~~110 111~~	~~02~~	~~02~~	~~02~~	~~02~~
0	1	0000 0001 1110 01	~~000 001~~ ~~010 011~~ ~~100 101~~ ~~110 111~~ 010	ab	bc	ab	bc
0	1	0000 0001 1011 01	~~000 001~~ ~~010 011~~ ~~100 101~~ ~~110 111~~ 110	dd	dd	dd	dd

b = 4
C = 32
B = 8
N = 8
S = $\frac{B}{N}$ = 1

O.B = 2
~~O.S~~
S.B = 0
T.B = 16 - 2 - 0 = 14

Hit % = $\frac{2}{13}$ x 100 = 15.38 %

Miss = 100 - 15.38 = 84.61 %

LRU

0	1	0000 0001 1111 00	000 001 010 011 100 101	e0	e0	0e	0e
0	1	0000 0001 0011 00	000 001 010 100	ac	bc	ac	bc

4]. Use 4-way associative cache to facilitate memory access for the memory sequence above. You should fill in the binary form of the hex values. Show the final contents of the cache in the table below, and compute the hit rate and miss rate. For cache write hits use write through and for cache write misses use write no-allocate. Design Cache and fill the Cache with the memory sequence given above. [15 Marks]

5]. Consider a machine with a byte addressable main memory of 64Kbyte with 8byte Block size. Assume that direct mapped cache consisting 32 lines. Write answer of the following questions [20 Marks]

a). How 64Kbyte Main memory address divided into tag, block/line number, and byte number

b). Into which cache line will each of the following addresses be stored?

00011 → 3_{16}

00110 → 6_{16}

00011 → 3_{16}

01010 → 10_{16}

00000 → 0_{16}

01001 → 9_{16}

a) b=8 C=32×8=256 B=32

Total Memory = 64kb

Address bits = 16

O.B = 3

N=1 → B=32

S.B = 5

T.B = 16-5-3=8

c) 1. 0001 1010 0001 000
2. 0001 1010 0001 001
3. 0001 1010 0001 010
4. 0001 1010 0001 011
5. 0001 1010 0001 100
6. 0001 1010 0001 101
7. 0001 1010 0001 110
8. 0001 1010 0001 111

d) 256 = Total C in bytes

c). Suppose the word [illegible] 0001 1010 0001 1010 is stored in the cache. What are the addresses stored along with it?

d) How many [illegible] can be stored in the cache?

6]. Assume [illegible] cache of size 32KB, each block having four 64-bit words. [illegible] physical address and uses byte addressing. Find the [illegible] 6z and 11 bits. [20 Marks]

- Main Memory Size 2^{64}
- Tag Bits 51
- Offset Bits 5
- Index Bits 8

C = 32 × 1024 = 32768

B = 4W = 32

S = C/B = 1024

N = 4

Sets = B/N = 256

O.B = $\log_2(32) = 5$

S.B = $\log_2(256) = 8$

TB = 64 - 5 - 8 = 51

Q4: b = 4
C = 32
B = 8
N = 4

$S = \frac{8}{4} = 2$

S.B = 1

O.B = 2

T.B = 16 - 1 - 2 = 13

Set	V	Tag	Data
0	1	0000 0000 0101 0	1b bc 14 bf
	1	0000 0000 01100	ab bc 1b bc
	0	—	
	0		
1	1	0000 0000 01100	ab bc ab bc
	1	0000 0001 11011	2b bc ab bb
	1	0000 0010 0100	ab bc ab bc
	0		

Assume a [illegible] cache of 256Byte and data access sequence given below where the memory [illegible] are shown in the table on the right. Assume 4byte addresses and a 4 byte [illegible] required to design a cache and calculate a Cache Hit and Miss %.

Operation	Address
Read	0x01
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]
Read	0x[illegible]



$b = 4B$

$C = 256$

$B = \frac{256}{4} = 64$

$N = 1$

$S = \frac{B}{N} = 64$

$O.B = \log_2(4) = 2$

$S.B = \log_2(64) = 6$

Tag Bits $= 32 - 2 - 6 = 24$

Hits: 1

Miss: 14

Hit rate: $\frac{1}{15} \times 100 = 6.66\%$

Miss rate: $100 - 6.66 = 93.33\%$

8]. A computer system employs a two-level cache hierarchy consisting of a Level 1 (L1) cache and a Level 2 (L2) cache. The system processes memory requests with the following characteristics:

- L1 cache: [illegible] memory requests, out of which 2550 requests result in a cache hit.
- L2 cache: [illegible] memory requests, out of which 2520 requests result in a cache hit.
- L1 cache: [illegible] nanosecond (ns)
- L2 cache: [illegible] ns
- Main memory: [illegible] (miss penalty): 100 ns

Calculate the following:

1. The number of memory accesses that result in a cache hit in the L1 cache.
2. The number of memory accesses that result in a cache hit in the L2 cache.
3. The number of memory accesses that result in a cache miss in both the L1 and L2 caches.
4. The effective access time (EAT) for this cache hierarchy.

1) Hit % $= \left(\frac{2550}{3000}\right) \times 100 = 85\%$

2) Hit % $= \left(\frac{2520}{2800}\right) \times 100 = 90\%$

3) L1: $3000 - 2550 = 450$

L2: $2800 - 2520 = 280$

Total $= 730$

4) $(0.85)(1) + (0.15)(9) + (0.1)(100) = 12.2$

Set	V	Tag	Data
0000	0		
0001	0		
0010	1	0000 0000 11	ab bc ab bc
0011	1	0000 0000 00	ab abc ab bc
0100	1	0000 0010 00	1c bb ab bb
0101	1	0000 0110 00	ab bc ab bc
0110	0		
0111	0		
1000	1	0000 0000 10	ab bb ab bb
1001	1	0000 0000 01	ab bc ab bc
1010	1	0000 0000 01	ab bc ab bc
1011	0		
1100	1	0000 0010 00	1b bb 1b be
1101	1	0000 0016 00	ff ff ff ff
1110	0		
1111	0		